AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 4)

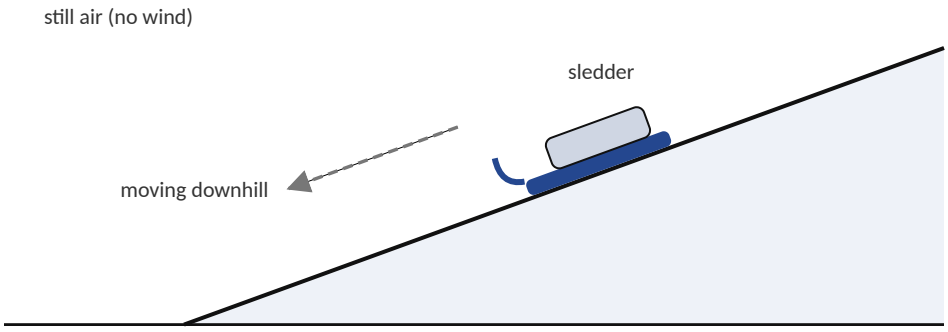
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

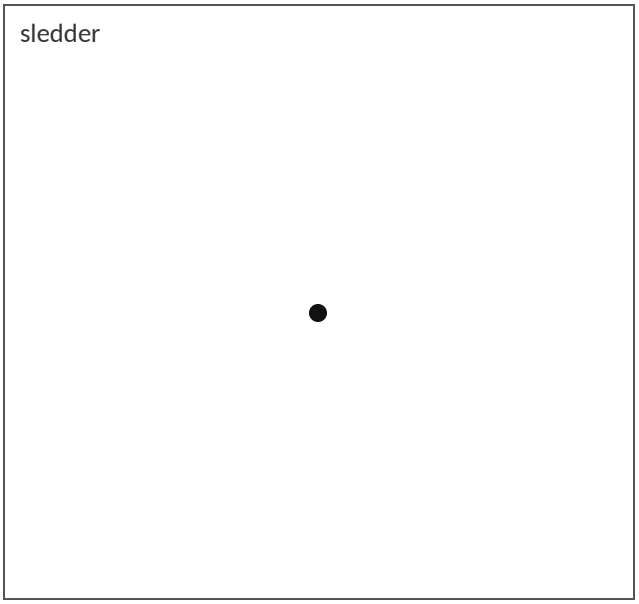
Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 50 points total.

1. (12 pts) Draw a free-body diagram for a kid on a cardboard sled sliding down a grassy hill. Friction and drag are **NOT** negligible. Label each vector you draw.





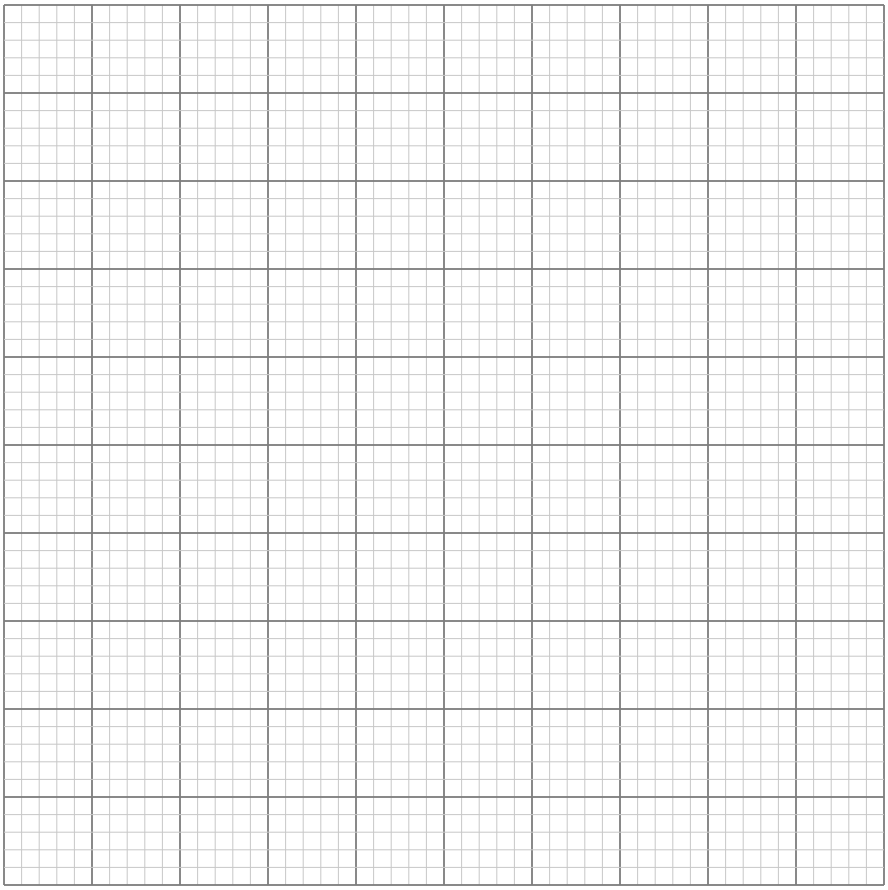
2. (14 pts) Students measure the drag force F_D on a speed skater in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
4.00	2.89	
6.00	5.76	
8.00	10.4	
10.0	17.4	
12.0	24.1	
14.0	34.0	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

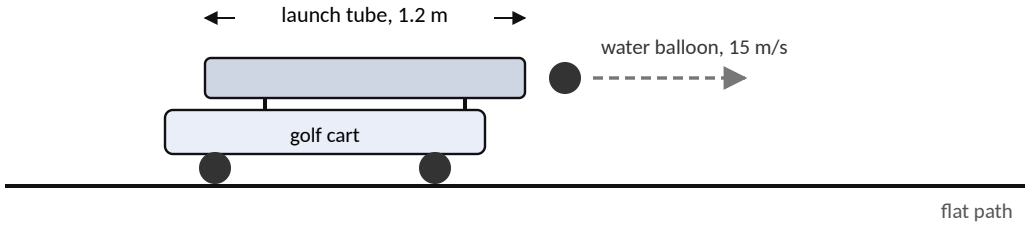
(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.

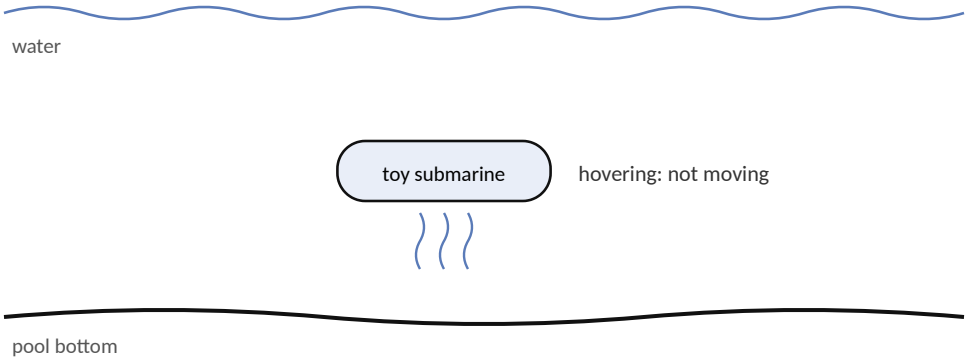


(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

3. (12 pts) A camp counselor launches a 0.45-kilogram water balloon horizontally from a launcher on a 520-kilogram golf cart. The water balloon experiences constant acceleration while inside the 1.2-meter-long launch tube, exiting at a speed of 15 meters per second. If the golf cart had its brakes off and if friction with the path were negligible, what would be the magnitude of the golf cart's acceleration?



4. (12 pts) A 620-gram toy submarine normally rests on the bottom of a pool. However, it can hover in the water by generating 0.90 newtons of upward thrust with its propeller. What is the magnitude of the buoyant force on the toy submarine?

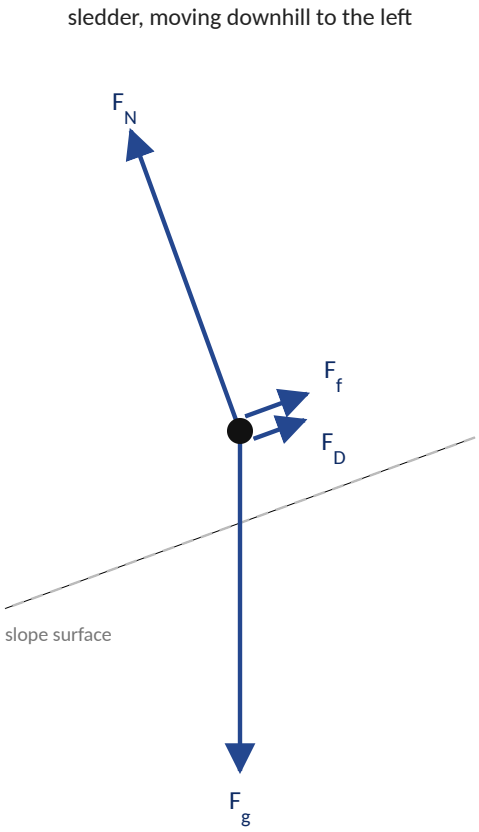


Answer Key – Practice Test, Version 4 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures.

Numerical problems: diagram first, then a = ΣF/m (equation sheet), written a = F_{net}/m. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagram (12 pts)



Step 1 **Every force on the sledder: exactly four.**

Earth pulls on it: F_g. The grass pushes on it two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on it: drag F_D.

Step 2 **Directions. No angle is needed.**

F_g: straight down, vertical.

F_N: perpendicular to the slope, out of the surface, so it leans toward the left.

F_f: along the slope, uphill (up and to the right), opposite the sliding.

F_D: opposite the motion through the air. In still air the sledder moves downhill through the air, so drag also points uphill, as its own arrow beside F_f.

Lengths are not graded: the motion (speeding up or not) is not stated. Every arrow starts on the dot and is labeled.

Watch out: no components of F_g drawn as extra arrows, no fifth "air resistance" arrow on top of drag, and no arrow pointing down the slope.

2. Drag linearization (14 pts)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

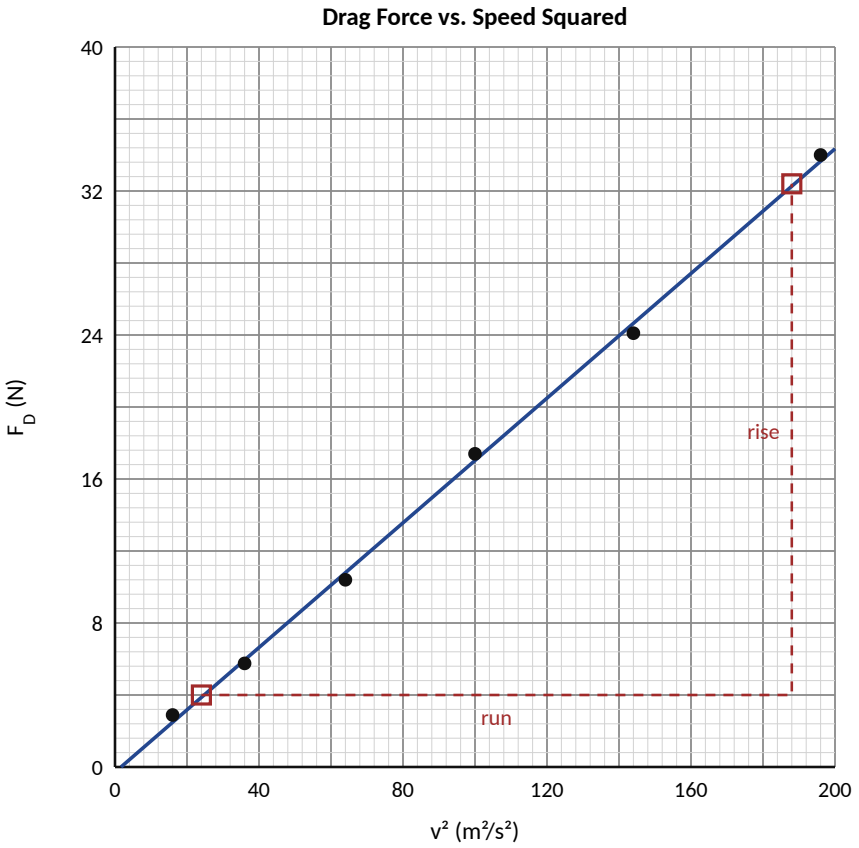
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(4.00\ m/s)^2 = 16.0\ m^2/s^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
4.00	2.89	16.0
6.00	5.76	36.0
8.00	10.4	64.0
10.0	17.4	100
12.0	24.1	144
14.0	34.0	196

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

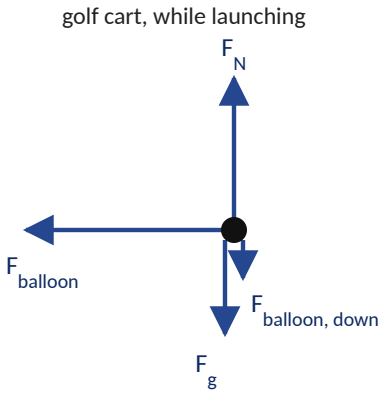
Points read off the line, away from data points: (24 m²/s², 4.0 N) and (188 m²/s², 32.4 N)

$$\text{slope} = \frac{32.4\ \text{N} - 4.0\ \text{N}}{188\ \text{m}^2/\text{s}^2 - 24\ \text{m}^2/\text{s}^2} = \frac{28.4\ \text{N}}{164\ \text{m}^2/\text{s}^2} = 0.17317\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.173\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.159 to 0.187 N·s²/m² from a reasonable line earns credit.

Units: N ÷ (m²/s²) = N·s²/m², and N = kg·m/s², so N·s²/m² = kg/m.

3. Recoil (12 pts)



Step 1 **Diagram of the golf cart while the water balloon is inside.**

Horizontal: only the water balloon pushing back on it ($F_{balloon}$). Vertical: F_N up, F_g down and the water balloon pressing down ($F_{balloon,\ down}$), all balanced, so they stay out of the horizontal equation. Not to scale.

Step 2 **Kinematics for the water balloon: no time given, so use #3.**

$v_0 = 0$, $v = 15$ m/s, $\Delta x = 1.2$ m (the launch tube length)

$v^2 = v_0^2 + 2a\Delta x \rightarrow a_{water\ balloon} = \frac{v^2}{2\Delta x} = \frac{(15\ m/s)^2}{2(1.2\ m)} =$

$93.75\ m/s^2 \approx$ **$94\ m/s^2$**

Step 3 **Second law on the water balloon, then the third law.**

$F_{net} = ma \rightarrow F = (0.45\ kg)(93.75\ m/s^2) = 42.19\ N \approx$ **$42\ N$**

$F_{water\ balloon\ on\ golf\ cart} = -F_{golf\ cart\ on\ water\ balloon}$: the same 42 N, backward.

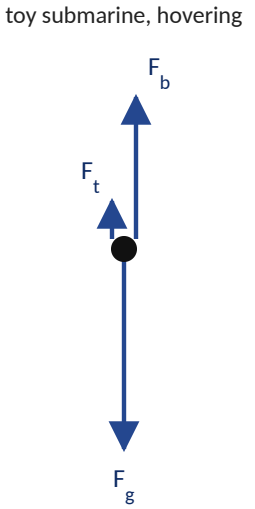
Step 4 **Second law on the golf cart.**

$a_{golf\ cart} = \frac{F}{M} = \frac{42.19\ N}{520\ kg} = 0.08113\ m/s^2 \rightarrow$ **$0.081\ m/s^2$, magnitude (it points backward)**

Using the launch tube length as the water balloon's distance is fine here: the golf cart moves back only about 0.1 cm while the water balloon is inside.

Watch out: equal and opposite forces, unequal accelerations: divide the same F by the much bigger mass.

4. Thrust and buoyancy (12 pts)



Step 1 **Diagram of the toy submarine while it hovers.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: it is off the bottom. Hovering means not moving, so $a = 0$ and the two up arrows together match F_g.

Step 2 **Newton's second law, up is +, letters first.**

$a = \frac{F_{net}}{m} \rightarrow F_b + F_t - mg = ma$, with $a = 0$

$F_b = mg - F_t$

Step 3 **Numbers, units in every parenthesis.**

$m = 620\ g = 0.62\ kg$

$F_b = (0.62\ kg)(9.8\ m/s^2) - 0.90\ N = 6.08\ N - 0.90\ N = 5.18\ N \rightarrow$ **$F_b = 5.2\ N$**

Watch out: thrust and buoyancy both point up. Writing $F_b - F_t - mg = ma$ (thrust down) flips the sign of the thrust.

1. Free-body diagram

Suggested partial credit	pts
F_g straight down, labeled	3
F_N perpendicular to the slope, out of the surface, labeled	3
F_f uphill along the slope, labeled	2
F_D uphill, opposite the motion, labeled, its own arrow	2
Arrows start on the dot; no components and no extra forces	2
Total	**12**

2. Drag linearization

Suggested partial credit	pts
(a) F_D (N) vertical, v^2 (m^2/s^2) horizontal	3
(a) Slope = K	1
(b) v^2 column correct, with units	2
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid; points plotted	1
(c) One straight best-fit line	1
(d) Two well-separated points read off the line	2
(d) K from 0.159 to 0.187, with units $N \cdot s^2/m^2$ (or kg/m)	2
Total	**14**

3. Recoil

Suggested partial credit	pts
#3 with $v_0 = 0$, solved for a in letters	2
$a = 94\ m/s^2$ for the water balloon	2
$F = ma = 42\ N$	3
Third law: same force, opposite direction, on the golf cart	2
$a = F/M = 0.081\ m/s^2$ (magnitude), units and sig figs	3
Total	**12**

4. Thrust and buoyancy

Suggested partial credit	pts
Diagram: F_g down, thrust and buoyancy up, no normal force	3
Hovering, so a = 0	2
$F_b + F_t - mg = ma$ with correct signs	3
Solved in letters: $F_b = mg - F_t$	1
Grams to kg; units in every parenthesis	2
$F_b = 5.2$ N	1
Total	**12**